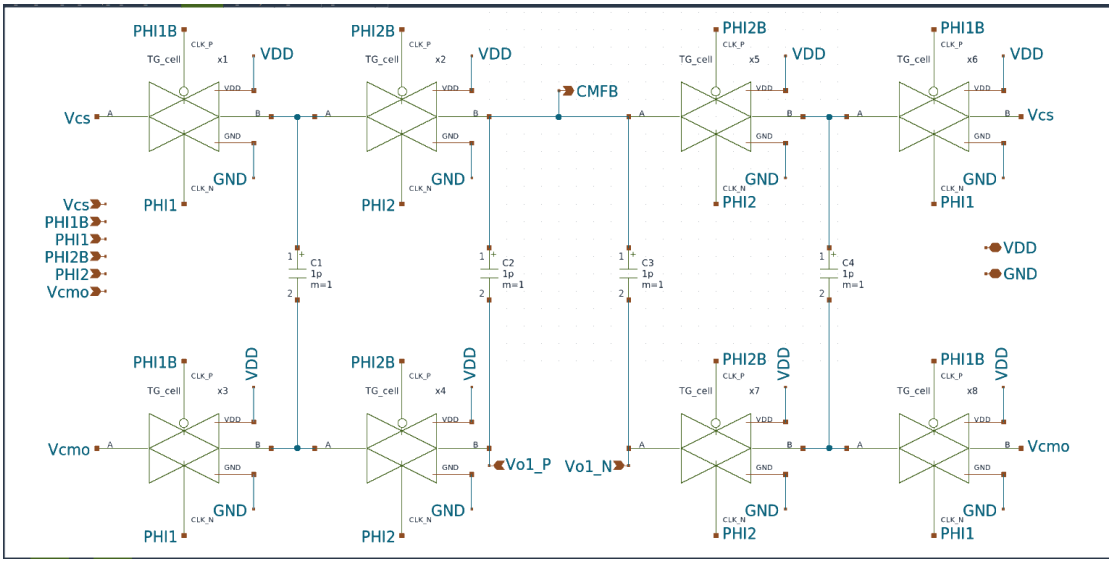
ENGINEERING DESIGN NOTE / BLOCK 02

From a Single Transmission Gate to a Working Feedback Loop

Design and Verification of an 8-TG Switched-Capacitor Common-Mode Feedback Network

Transistor-level implementation, OTA integration, and first PVT lessons



Evidence status: nominal closed-loop operation demonstrated; fixed-bias PVT robustness and 4-MHz timing remain open.

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Abstract

A transmission gate may look like the smallest possible starting point for an analog design. In this project it became the reusable unit of a much larger system: an eight-switch, four-capacitor switched-capacitor common-mode feedback (SC-CMFB) network for a fully differential operational transconductance amplifier (OTA). The work progressed through transistor-level switch design, symbol reuse, standalone validation, first-stage node identification, broken-loop integration, and closed-loop transient analysis. A decisive debugging step was to distinguish the OTA's first-stage common-mode nodes from its final output nodes. A three-point DC sweep measured an approximately $-118.54\,\mathrm{V/V}$ first-stage control slope, supporting the intended negative feedback direction. At TT, 1.8 V, and 27 °C, the 100 µs closed-loop run produced a first-stage mean of 0.930 101 V against a 0.929 274 2 V reference, with approximately 0.470 mV peak-to-peak ripple. The CMFB control mean was 0.900 678 V. Seven initial PVT tests then revealed a substantial limitation: fixed external bias voltages do not maintain the desired operating point at supply and temperature extremes. The result is a functional nominal prototype, accompanied by an explicit path toward more robust biasing and faster-clock verification.

Keywords: transmission gate; switched-capacitor; common-mode feedback; fully differential OTA; SKY130; PVT; mixed-signal verification.

1 Why the work began with one switch

The wider portfolio targets a second-order switched-capacitor delta-sigma modulator. In that setting, an OTA must amplify the differential signal without allowing its internal common-mode level to drift freely. The first instinct might be to add a feedback block around the amplifier and move on. We chose a more transparent route: begin with the switch itself, understand when it conducts, and only then build the network that performs the feedback action.

The two first-stage OTA outputs are written as $V_{\mathrm{O1,A}}$ and $V_{\mathrm{O1,B}}$. Their common- and differential-mode components are

$$V_{\mathrm{O1,CM}} = \frac{V_{\mathrm{O1,A}} + V_{\mathrm{O1,B}}}{2}, \qquad V_{\mathrm{O1,D}} = V_{\mathrm{O1,A}} - V_{\mathrm{O1,B}}. \tag{1}$$

The final OTA outputs require separate notation:

$$V_{\mathrm{out,CM}} = \frac{V_{\mathrm{out,p}} + V_{\mathrm{out,n}}}{2}. \tag{2}$$

The distinction is not cosmetic. This specific SC-CMFB network senses the *first-stage* nodes; the final output common mode is a downstream observation, not the correct sensing input for this feedback network.

The SC-CMFB's calibrated first-stage reference is $V_{\mathrm{O1,ref}} = 0.929\,274\,2\,\mathrm{V}$. It is intentionally *not* identical to the nominal final-output level near 0.9 V. The reference was obtained from a nominal OTA sweep, not simply assumed to be $V_{DD}/2$.

2 Designing a reusable transmission gate

2.1 A parallel NMOS–PMOS signal path

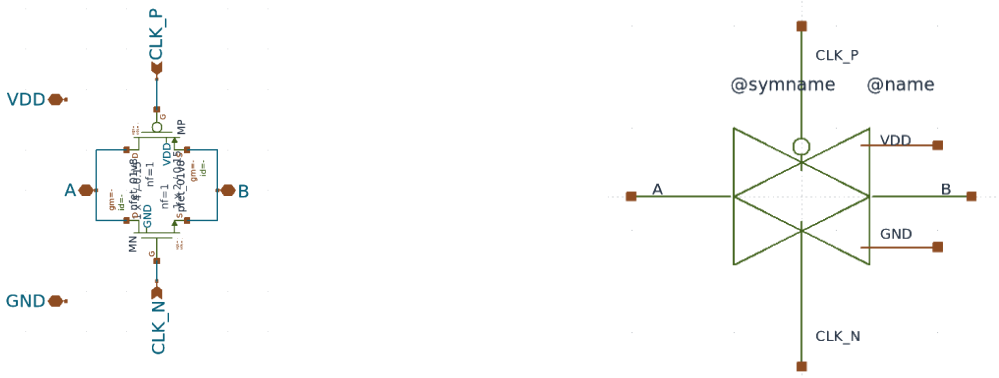
One NMOS and one PMOS form the switch. Their signal terminals are connected in parallel, while their gates receive complementary clocks. The nominal SKY130 1.8-V device sizing

used in the TG cell was $W_n/L_n = 2/0.15$ and $W_p/L_p = 4/0.15$, with geometry expressed in µm. These dimensions document the cell used for this design; they are not a claim that it has been layout-optimized.

When $\phi = V_{DD}$ and $\overline{\phi} = 0$, both devices are commanded ON. When the levels are reversed, both are commanded OFF. To first order, their combined ON resistance is

$$R_{\text{on,TG}} \simeq \left(\frac{1}{R_{\text{on,n}}} + \frac{1}{R_{\text{on,p}}} \right)^{-1}. \tag{3}$$

The actual resistance is signal-dependent because V_{GS}, threshold voltage, and body effect change during switching. The PMOS improves conduction toward the upper signal rail, while the NMOS helps near the lower rail.



(a) Transistor-level TG cell. **(b)** Reusable Xschem symbol.

Figure 1: The design starts with one physical switch and turns it into a modular building block. The images are print-friendly versions of the supplied Xschem screenshots.

2.2 What the switch must accomplish during each phase

A switched-capacitor stage needs time to settle after a TG turns ON. For an approximate lumped RC path,

$$|V_C(t) - V_\infty| = |V_C(0) - V_\infty| \exp\left(-\frac{t}{R_{\text{eq}} C_{\text{eq}}} \right), \quad t_{\text{on}} \geq R_{\text{eq}} C_{\text{eq}} \ln \frac{1}{\epsilon}. \tag{4}$$

For example, with a nominal 1 pF storage capacitor, 300 ns available ON time, and a 0.1% residual-error target, the simple RC model requires $R_{\text{eq}} \lesssim 43.4\,\text{k}\Omega$. This is a *screening estimate*, not the measured TG resistance; extra parasitics and nonlinear switch behaviour are omitted.

The individual TG was first subjected to clock and signal-transfer tests before being cloned. This order mattered: a wiring or polarity error in one symbol would otherwise have been replicated eight times in the final network. ON resistance, leakage over full PVT, and charge injection still require dedicated quantitative sign-off.

3 From one TG to the eight-switch SC-CMFB

3.1 Four capacitors, two active phases

The verified TG symbol was instantiated eight times and connected with four nominal 1 pF capacitors. The Xschem implementation uses the clocks $\phi_1, \overline{\phi}_1, \phi_2, \overline{\phi}_2$. Its main external nodes are V_{cs}, V_{cmo}, the two first-stage sensed outputs, and the CMFB control output.

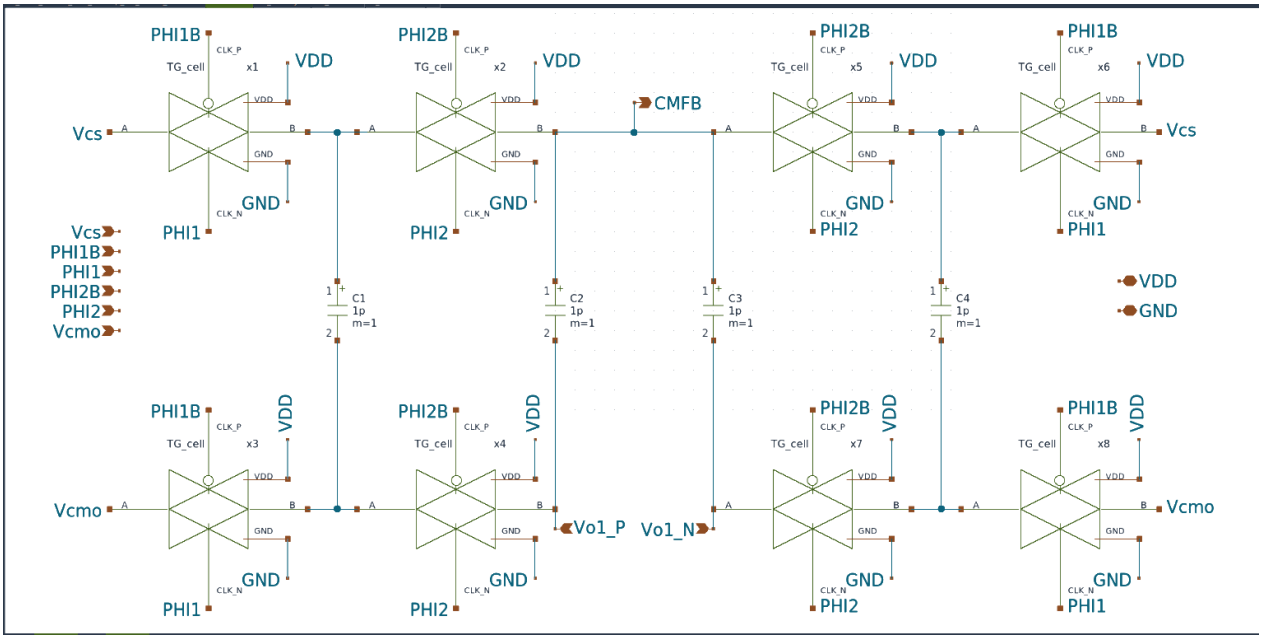


Figure 2: Implemented eight-TG / four-capacitor SC-CMFB schematic (Xschem). The left and right halves sample and redistribute charge from the two OTA first-stage outputs.

The standalone stimulus uses 1 µs clock period, with ϕ_1 active approximately from 100 ns to 402 ns, and ϕ_2 from 600 ns to 902 ns. The inactive interval between phases provides approximately 198 ns of dead time. The barred waveforms are the corresponding complementary gate controls; the "b" in `PHI1B` and `PHI2B` means *bar*, or logical complement.

The charge on each storage capacitor follows $Q_i = C_i V_i$. In a lossless redistribution step, total charge at an isolated node is conserved:

$$\sum_i C_i V_i^{\text{before}} = \sum_i C_i V_i^{\text{after}}. \tag{5}$$

For the symmetric, equal-capacitor configuration, simulations approximately follow the common-mode transfer relation

$$V_{\text{CMFB}} \approx V_{cs} + (V_{\text{O1,CM}} - V_{cmo}) + \varepsilon_{\text{SC}}, \tag{6}$$

where ε_{SC} collects finite settling, charge injection, leakage, and other non-idealities. Equation (6) is an *empirical approximation supported by the observed operating points*, not an exact transfer function extracted from a full symbolic nodal-charge derivation.

At nominal temperature, the simplified sample-noise scale of one 1 pF capacitor is

$$v_{n,\text{rms}} \approx \sqrt{\frac{kT}{C}} = \sqrt{\frac{(1.380649 \times 10^{-23})(300)}{10^{-12}}} \approx 64.4\,\mu\text{V}. \tag{7}$$

This is a per-capacitor kT/C estimate, not the complete differential or output-referred noise of the four-capacitor feedback network.

3.2 Why we validated the network alone first

A standalone testbench applied the reference voltages, phase clocks, and first-stage input pair to the SC-CMFB. Positive, negative, and zero-differential input combinations were explored. The block produced a bounded control voltage with phase-dependent switching ripple and approached a repeatable value. This established basic functionality without mixing switch debugging with OTA bias debugging.

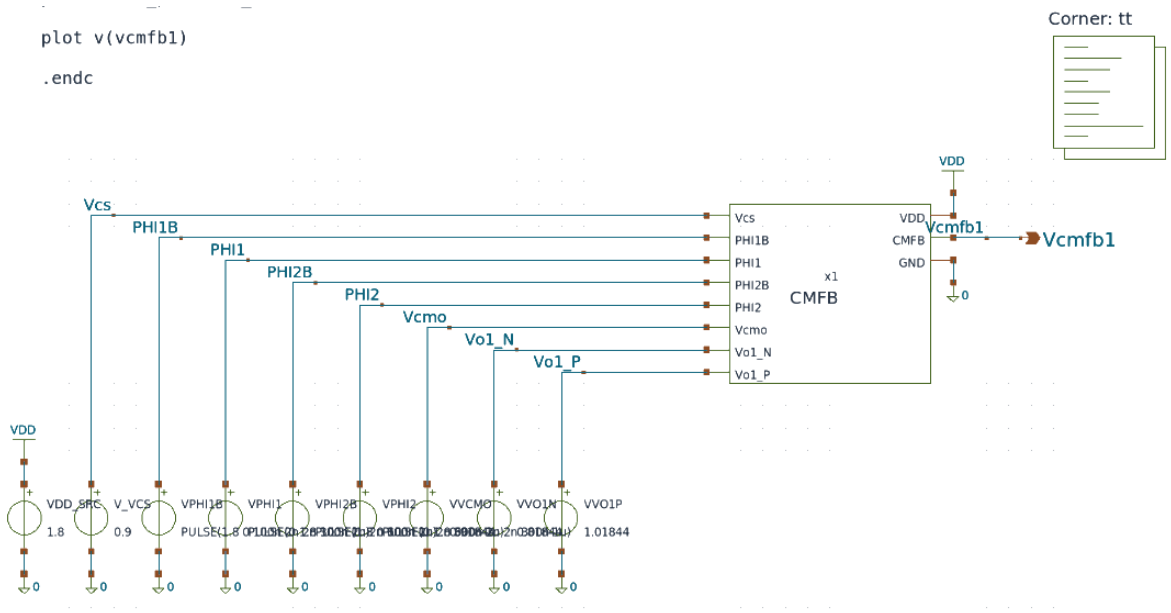


Figure 3: Standalone SC-CMFB testbench showing the four clock inputs, reference sources, sensed first-stage voltage sources, and control output.

The separate test is not a proof of closed-loop stability. Its value is narrower and more useful: it verifies that the generated control signal changes as the sensed common-mode changes, and that the intended TG phases are present.

4 Making SC-CMFB part of a real OTA loop

4.1 The most instructive mistake: sensing the wrong stage

Our first integration attempts observed the final OTA outputs. The resulting control trend looked inconsistent, and several simulations also exposed a netlist naming error, a floating control measurement, an unsupplied CMFB pin, and swapped node labels. Rather than tune transistors around those symptoms, we inspected the netlist and separated the OTA's internal first-stage pair from its final outputs.

A local three-point sweep of the externally driven OTA CMFB gate resolved the sign unambiguously:

V_{cmfb1} (V)	$V_{\mathrm{O1,CM}}$ (V)	$V_{\mathrm{out,CM}}$ (V)
0.8998	0.953044	0.378690
0.9000	0.929274	0.927668
0.9002	0.905628	1.351370

The central-difference slope of the *first-stage* common mode is

$$S_{\mathrm{OTA}} \approx \frac{0.905628 - 0.953044}{0.9002 - 0.8998} = \boxed{-118.54\ \mathrm{V/V}}. \tag{8}$$

The final-output slope over the same narrow interval has the opposite sign, approximately $+2.43 \times 10^3$ V/V. This explains why a feedback connection based on the final nodes was misleading. The earlier standalone operating-point solver also produced a singular-node warning; the three converged DC-sweep points, rather than the failed OP values, were used for this calculation.

4.2 A broken-loop handoff before closing the feedback path

We connected the actual first-stage nodes to SC-CMFB but kept the OTA's control gate at an independent 0.900 000 V source. In this configuration, the sensed first-stage common mode averaged approximately 0.929 257 V, while the SC-CMFB generated about 0.898 592 V. The approximately 1.41 mV difference between the expected control setting and the emitted control level gave us a clear interface metric before physically closing the loop.

Once the generated CMFB output was routed back to the OTA control input, the independent source was removed. The successful integration required the CMFB VDD pin to connect to the real supply and the feedback net to have one consistent node name. These were netlist-level details, but they determined whether the waveform could be trusted.

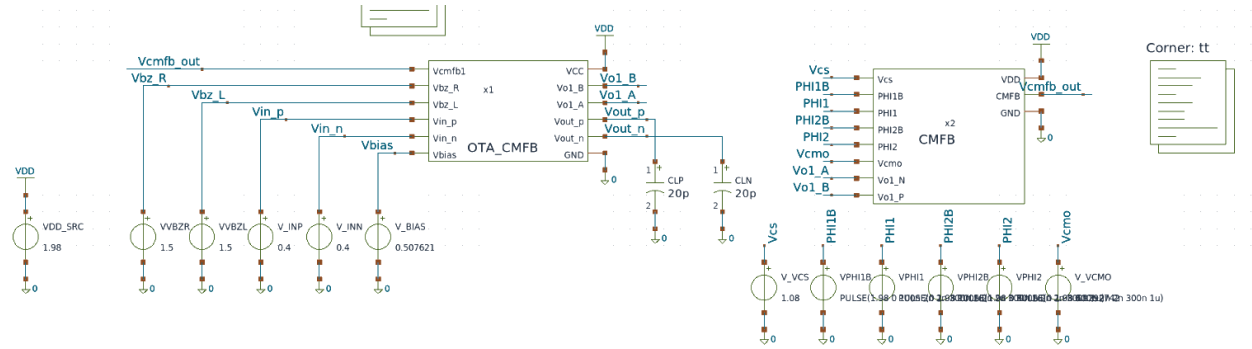


Figure 4: Transistor-level OTA–SC-CMFB closed-loop testbench. The first-stage nodes are the sensed pair; the CMFB output returns to the OTA control gate. The displayed screenshot is a bias-characterization variant at elevated VDD, not the source of the nominal result table.

4.3 The 100 µs nominal closed-loop result

At TT / 1.80 V / 27 °C, the settled measurement window was 90 µs to 100 µs. Both differential inputs were held at 0.4 V and the final outputs were loaded by 20 pF each. Thus the near-zero differential residue is an expected balance check, *not* evidence of differential gain or dynamic signal fidelity. The recorded values are:

Table 1: Nominal closed-loop transient measurements; Xschem/ngspice, last 10 µs.

Metric	Mean / error	Peak-to-peak ripple
First-stage common mode, $V_{O1,CM}$	0.930 101 4 V	0.4696 mV
First-stage target, $V_{O1,ref}$	0.929 274 2 V	—
First-stage target error	0.8272 mV	—
CMFB control, $V_{cmfb,out}$	0.900 677 6 V	0.6440 mV
Final output common mode, $V_{out,CM}$	0.900 133 2 V	5.6199 mV
Mean output differential residue	near 0 V	—

The reference error is $e_{CM} = 0.9301014 - 0.9292742 = 0.8272$ mV. In the last two 10 µs windows, changes in mean first-stage and control voltages were approximately -1.2 µV and -1.1 µV, respectively. This is good evidence of a nearly periodic nominal operating state. It does *not* independently establish sampled-data loop phase margin, recovery under a disturbance, or adequacy at the project's 4 MHz sampling target.

5 PVT screening: where nominal success stops being enough

We preserved the nominal external biases at $V_{bias} = 0.507\,621\,\text{V}$, $V_{bz,L} = V_{bz,R} = 1.5\,\text{V}$, $V_{cs} = 0.9\,\text{V}$, and $V_{cmo} = 0.929\,274\,2\,\text{V}$ while varying one external condition at a time. The clock HIGH level followed VDD for voltage sweeps. This was deliberately a *fixed-bias stress test*, not the characterization of a finished bias generator.

Table 2: Seven initial closed-loop PVT conditions; a nominal-target error is reported, not a process qualification verdict.

Condition	$V_{O1,CM}$ (V)	Error (mV)	V_{CMFB} (V)	$V_{out,CM}$ (V)
TT, 1.80 V, 27 °C	.930101	+.827	.900678	.900133
TT, 1.80 V, □40 °C	.878113	-51.161	.848921	1.504754
TT, 1.80 V, 125 °C	1.020898	+91.624	.991365	.312116
TT, 1.62 V, 27 °C	.749679	-179.595	.721616	1.455200
TT, 1.98 V, 27 °C	1.109066	+179.792	1.078414	.288498
SS, 1.80 V, 27 °C	.938441	+9.167	.908919	.898089
FF, 1.80 V, 27 °C	.921904	-7.370	.892594	.899302

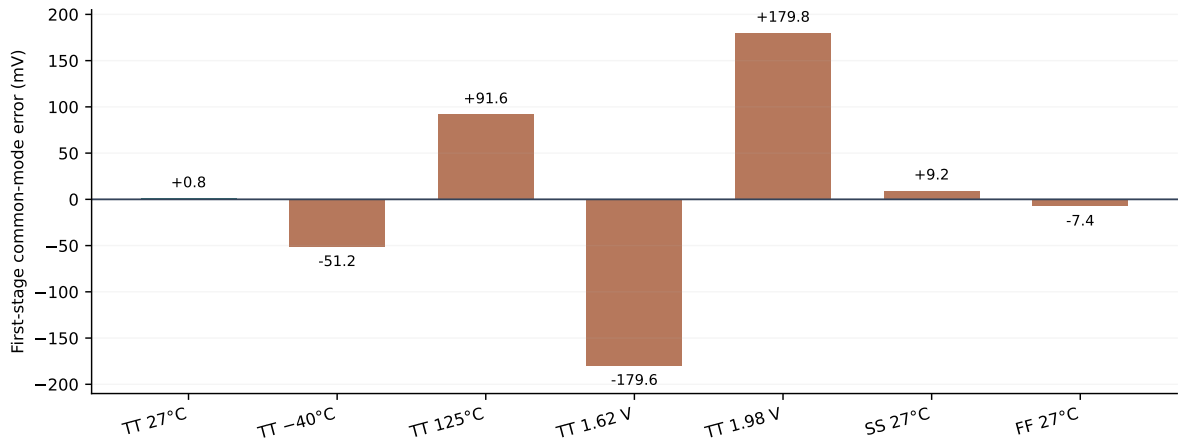


Figure 5: First-stage common-mode error across the seven simulated process, supply, and temperature conditions. Values are extracted from the supplied simulation screenshots.

The pattern is unusually instructive. Process-only SS and FF shifts are moderate relative to the supply excursions, while the two supply corners shift the sensed common mode by approximately $\pm 180\,\text{mV}$. At low supply the final OTA output approached the upper rail; at high supply it fell toward the lower rail. Nearly zero drift at a bad operating point is not a successful regulation result. It means the circuit has settled *somewhere*, not necessarily where the amplifier remains useful.

Importantly, Eq. (6) continues to describe the observed trend to first order. This suggests that the SC network still translates common-mode error into a control voltage, while the OTA and its fixed bias conditions lose their desired equilibrium. The evidence does not yet isolate whether current redistribution, device operating region, available control range, or more than one of these dominates.

The 1.98 V stress point should also be checked against the reliability limits of the specific SKY130 device models and PDK rules before it is treated as a valid continuous-operation corner.

6 The first bias experiment: changing V_{cs}

Instead of immediately resizing M13, the input pair, or the feedback capacitors, we changed the CMFB reference source. An automated transient sweep included V_{cs} values from 0.70 V to 1.10 V at three supplies. The terminal screenshots made available for this report show the final 1.10 V sweep point; the seven-point raw arrays have not yet been archived in a machine-readable table.

Table 3: Observed first-stage mean at two bias settings, TT and 27 °C.

VDD (V)	$V_{O1,CM}$, $V_{cs} = 0.90$ V	$V_{O1,CM}$, $V_{cs} = 1.10$ V	Change (mV)
1.62	0.749679	0.550443	-199.236
1.80	0.930101	0.730138	-199.963
1.98	1.109066	0.911039	-198.027

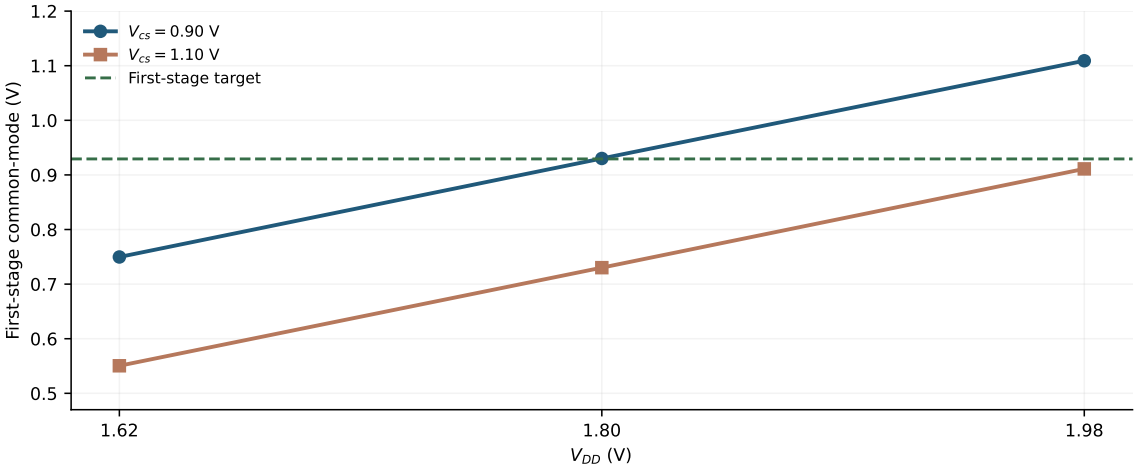


Figure 6: Shift in first-stage common mode when V_{cs} is increased from 0.90 V to 1.10 V; each marker corresponds to a measured endpoint, not to a full fitted transfer curve.

For these two endpoints, $\Delta V_{O1,CM}/\Delta V_{cs} \approx -1$ V/V. A linear interpolation suggests candidate reference values around 0.72 V, 0.90 V, and 1.08 V for VDD of 1.62 V, 1.80 V, and 1.98 V, respectively. These are *hypotheses for further transient and operating-region verification*, not optimized silicon-ready biases. Matching $V_{O1,CM}$ alone cannot guarantee final-output swing, device headroom, stability, or low noise.

7 What we can claim now, and what remains open

This block has reached a concrete milestone: a TG moved from a two-transistor cell to a reusable symbol; eight such symbols and four capacitors formed an SC-CMFB network; standalone behaviour was checked; the correct OTA first-stage nodes were identified; broken-loop testing clarified the interface; and a closed-loop nominal operating state was obtained. These are actual schematic-level steps, not a block diagram standing in for a circuit.

Still-open work consists of (i) transistor operating-region and current audits across PVT, (ii) a realizable supply/temperature-tracking bias architecture, (iii) differential stimulus and CM disturbance-recovery tests, (iv) sampled-data stability analysis and clock-feedthrough/noise checks, (v) TG charge injection and mismatch evaluation, and (vi) retiming the SC-CMFB from the demonstrated 1 MHz clocks to the intended 4 MHz system clock and rechecking settling. The present work is *nominally functional and PVT-characterized*, not silicon-signed-

off.

References and evidence notes

1. S. Pavan, R. Schreier, and G. C. Temes, *Understanding Delta-Sigma Data Converters*, 2nd ed., Wiley/IEEE Press, 2017, chap. 7. https://doi.org/10.1002/9781119258308.ch7.
2. Original Xschem schematics and ngspice terminal outputs supplied during the Block 2 design iterations, October 2026. All numbered circuit results in this document are reported from those screenshots; no claim of independent ngspice replication is made.
3. All schematic figures derive from the supplied Xschem screenshots. Their light-background variants improve print legibility; the connectivity has not been redrawn or changed. Charts are regenerated from tabulated simulator values, not from extracted waveform samples.